



Propagation Channel Scenarios (5G)



1. Definition

- Propagation channel scenarios refer to **different real-world environments in which wireless signals travel from transmitter to receiver.**
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Simple Meaning

👉 Signal behaves differently depending on location:

- City 🏙️
 - Village 🌾
 - Inside building 🏠
 - Moving vehicle 🚗
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Basic Diagram



→→→ Environment →→→ 📶
(Urban / Rural / Indoor / Mobility)



2. Types of Propagation Scenarios



1. Urban Scenario (City Area)



Description

- Area with **many buildings, towers, obstacles**

Characteristics

- Strong **reflection**
- Severe **multipath propagation**
- High **interference**
- Signal blockage

Diagram

 →   → 
Reflection & blockage

Example

- Signal in Hyderabad city between tall buildings

2. Rural Scenario (Village Area)

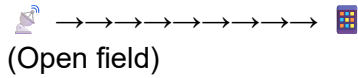
Description

- **Open areas**, fewer obstacles

Characteristics

- Long-distance communication
- Less interference
- Mostly **Line-of-Sight (LOS)**

Diagram



Example

- Signal in villages or highways
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3. Indoor Scenario

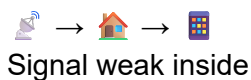
Description

- Communication **inside buildings**
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Characteristics

- Signal blocked by walls
 - High **attenuation**
 - Reflection dominant
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Diagram



Example

- Weak signal inside room
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4. High Mobility Scenario

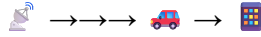
Description

- User is moving fast (car/train)
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Characteristics

- **Doppler effect**
 - Rapid signal changes
 - Frequent handover
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Diagram


Moving user

Example

- Using phone in train
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5. Dense Urban / Small Cell Scenario (Advanced)

Description

- Highly crowded areas
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Characteristics

- Very high user density
- Small cell deployment

- High data demand
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Example

- Stadium, malls
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◆ 3. Comparison of Scenarios

Scenario	Obstacles	Range	Interference
Urban	High	Short	High
Rural	Low	Long	Low
Indoor	Medium	Short	Medium
Mobility	Variable	Medium	High

◆ 4. Importance of Studying Scenarios

- Helps design better networks
- Improves signal quality
- Helps in planning base stations
- Reduces interference